

Lesson 16 — Build the Model & Storing Data

Matches LEGO Teacher Portal lesson: "Save or Score".

Objective (I can...): I can build the variables model, create a clearly named variable, store a value, and read it. **CS concept:** Variables as named storage; naming and data values · **Builds on:** Lesson 15 (decisions we can now base on stored data).

AI Connection (2 min): A model stores everything it "knows" as numbers in memory — weights, counts, scores. *First-principles question:* "What's the difference between a program that *remembers* across runs and one that starts fresh every time?"

Materials: LEGO CS & AI kit + device per pair, notebooks. **This is the unit's build lesson** — build the Variables model now and keep it assembled through Lesson 20.

60-minute flow - Warm-up (8): "Name it well" — coach shows poorly named vs. clearly named variables ("x" vs. "scoreRed"); pairs rank them and state a naming rule. Good names are documentation. - **Concept + demo (10):** Create a variable, set it to a starting value, display it on screen, then change it and watch the display update. Name the difference between a value and the box that holds it. - **Build/Code (30):** Pairs **build the Variables model** (the unit's one build), then create a clearly named variable, set a starting value, and display it. They record the variable's name and purpose in the notebook. *Swap at 15 min.* - **Challenge (7):** Set a different starting value and re-run; predict the displayed value first. - **Share + pack-up (5):** Pairs show their named variable and its value and justify the name. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator names the variable and reads its value. Swap at 15 min. (*Trio rotates.*)

Differentiation: *Support* — a worked example with a well-named variable to adapt. *Extension / fast-finisher* — create two variables of different purposes. *Reteach* — write the value on paper, then mirror it on screen so "stored value" is concrete.

Exit check: Pair creates a clearly named variable, stores and displays a value, and can justify the name.

At-home (optional): Name five variables a video game would need to track and say what each holds. ~10 min.